



SCORCHED MILE



Artificial
Intelligence in
Games
Tournament

PROD323-26S1: Game Engine and AI

Assignment 1 – AIGT Racing Royale

Total Assessment = 20% (30 points)

**This assignment is worth 20% (max cap) of the final grade.
All submissions must be completed by Monday, 20 April 2025.**

Behaviour Design = 10% (15 points)

**The complexity and sophistication of the AI behaviour to
interact with the other competitor AIs and the environment.**

Character Design = 4% (6 points)

**The quality of your character movement mechanics and its
ability to navigate different terrains.**

Level Design = 6% (9 points)

**The creativity in designing the challenges of the game level for
AI racing and the effort in integrating multiple components.**

Tournament Bonus = 2% (3 points)

**The level of success of your AI agent at competing with the
other agents in different game levels.**

Marking Criteria:

Requirements		Marking Criteria					
		Abysmal (0%)	Subpar (20%)	Mediocre (40%)	Commendable (60%)	Superb (80%)	Epic (100%)
Behaviour Design	Terrain Navigation	Void	Standard approach	Slight heuristic tweak	Commendable heuristic	Novel heuristic	Extremely novel heuristic
	Sophistication	Void	Minimal illustration of intelligent behaviour	Acceptable level of intelligent behaviour	Good level of intelligent behaviour	Very high level of intelligent behaviour	Intricate behaviour with sophisticated actions/reactions
Character Design	Movement Mechanics	Void	Typical mechanics	Average mechanics	Good mechanics	Superb mechanics	Epically designed mechanics
	Adaptivity	Void	Minimal demonstration of adaptation	Somewhat adaptive to different conditions	Reasonably adaptive to different conditions	Highly adaptive to different conditions	Extremely adaptive to different conditions
Level Design	Creativity	Void	Minimal illustration	Acceptable effort	Good effort	Great effort	Masterpiece!
	Challenges	Void	Rather easy	Some level of challenge	Good level of challenge	Very challenging	Extremely challenging but possible to win
Tournament Performance	Champion (# Rank)	Void	3 rd overall	2 nd overall	1 st overall	-	-

Prizes

AGT Supreme Winner – The tournament winner(s) will be individually rewarded \$20 Steam gift card.

2nd and 3rd Places – Members will be individually rewarded a box of chocolate.

Best Level Design / Most Impressive Character / Most Cunning Behaviour – Each winner will be rewarded a bag of candies.

Rules (version 1.1 – subject to minor changes):

10/04/2026

B3: Clarified ForceType. B6: Updated force to 150 units. B7: Updated torque to 50 units. B9: Clarified casting.

20/03/2026:

A2: Clarified traps destruction – D4, D6: Clarified Obstacles – Added Section E: Combat and moved combat related points there.

Golden Rules

If it feels like cheating, it probably is

A. General Agreement

- A1. No instantiation
- A2. No “Destroy” function or any form of destruction
 - Except for **triggered traps**
- A3. No direct manipulation of transforms (position, rotation, and scaling)
- A4. No flying/hovering
- A5. No CPU-intensive operation critically affects the performance of the simulation. Generally, the framerate during the simulation of your agent navigating in your own level must average > 100 fps
- A6. All measurements are based on Unity default metrics. 1 unit = 1 meter.
- A7. Must not use Unity's NavMesh nor Asset Store A* algorithm
- A8. Over 50% of your starting mass must reach the finish line to complete the race.
- A9. All assets required for your agent should be under **Assets/_StudentFiles/YourUserCode/**. An exception is provided assets
- A10. Your agent's prefab root must be named your usercode (e.g. ABC123)

B. Behaviour Design

- B1. All movements are done through rigidbody
- B2. All movements must be driven by setting velocity or force and not position.
- B3. If using AddForce, you must use ForceType.Force and not Acceleration
- B4. Max linear velocity is 10 units
- B5. Max angular velocity is 7 units (by default)
- B6. Max force is 150 units
- B7. Max torque is 50 units
- B8. Max # raycast (line): 3
- B9. You may only use RayCast, you cannot use SphereCast, BoxCast, or CapsuleCast

C. Character Design

- C1. Must use one of the preset vehicle assets as the foundation of your agent's vehicle
- C2. Character dimension: w/d/h = 1-1.5 unit
- C3. The root GameObject of your character must be tagged "Agent"
- C4. Total mass = 50 units
- C5. Must have a rigidbody on the root of your GameObject.
- C6. Must have colliders. These cannot be enabled/disabled.
- C7. Max # colliders = 5; primitives only, no mesh colliders
- C8. Max particle system: 2
- C9. The only collision detection mode allowed is Discrete
- C10. Max # of particles per particle system: 100

D. Level Design

- D1. Terrain dimension: w/d/h = 128/128/32
- D2. Heightmap Resolution (Texture): 513, 513
- D3. Obstacles must be tagged "Obstacle".
- D4. Obstacles are placed in the world during level design
- D5. Obstacles may be dynamic and use collision to trigger them.
- D6. Obstacles may use animation for movement
- D7. Obstacles may not apply forces to agents.
- D8. The start line (prefab) must be added to your level. You can edit the look of it, but the functionality must be unchanged.
- D9. The start line must have room for up to 30 racers.
- D10. The finish line (prefab) must be added to your level. You can edit the look of it, but the functionality must be unchanged.
- D11. Theme: Post-Apocalypse
- D12. Must have background music

E. Combat

E1. Melee is allowed.

- You may add force directly to collided agents
- Must use `rigidbody.AddForce()`
- Maximum Force: **200**
- Allowed ForceType: **Impulse**

E2. Traps are acceptable

E3. Traps are placed by the agent (e.g. dropped behind)

- Must be placed no more than a reasonable distance from the agent

E4. Traps may not move after deployment

E5. Maximum number of traps = **4**

E6. Traps must be added to the agent prefab before the race (disabled)

E7. Maximum trap radius: **0.5f**

E8. Traps must be tagged "**Obstacle**"

E9. Must be one of the given trap types below:

E10. Explosive Traps:

- Must use a **Trigger**
- Must use `rigidbody.AddExplosiveForce()`
- Maximum Force: **500**
- Maximum Upwards force: **3**
- Allowed ForceType: **Impulse**
- Must be destroyed as soon as its triggered

E11. Spin Traps:

- Must use a **trigger**
- Must use `rigidbody.AddTorque()`
- Can only add torque on the **y-axis** (`Vector3.up * torqueStrength`)
- Maximum torque: **250**
- Allowed ForceType: **Impulse**

E12. Physical Traps

- May be knocked around by agents (e.g. not kinematic)
- Must **not** be a trigger
- Must **not** apply force/torque